

ADITI SAXENA

Doctoral Researcher | PEM Fuel Cells, Electrolysers & Hydrogen Technology

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RESEARCH PROFILE

Doctoral researcher specializing in the mechanical and electro-mechanical reliability of polymer electrolyte fuel cell components, including membrane electrode assemblies (MEAs) and catalyst coated membranes (CCMs), combining experimental micromechanics with statistical failure analysis. Experimental track record at Forschungszentrum Jülich in nanoindentation, microscratch testing, and a custom electro-mechanical puncture methodology developed during the doctoral work, supported by Python-based statistical pipelines including OLS regression, F-statistics, and Weibull maximum likelihood estimation with right-censored data. Two first-author manuscripts in preparation. Seeking a role on mechanical and electrochemical durability of polymer-based energy systems, applying micromechanical testing and statistical failure analysis to membrane-interface reliability in fuel cells, electrolysers, and related electrochemical devices.

PUBLICATIONS

Manuscripts in preparation

- **Saxena, A.**, Utsch, N., Brinckmann, S., Scheepers, F., & Schwaiger, R. *Electro-Mechanical Puncture of Hydrated PEMFC Membranes: Quantifying Pin-hole Initiation under Applied Bias*. Manuscript in preparation.
- **Saxena, A.**, Utsch, N., Brinckmann, S., Scheepers, F., & Schwaiger, R. *Micromechanical failure analysis of thin-film electrodes on polymer membranes for fuel cells*. Manuscript in preparation.

RESEARCH EXPERIENCE

Doctoral Researcher | HITEC Fellow

Sep 2022 – Present (Expected: Oct 2026)

Forschungszentrum Jülich GmbH – Institute of Energy Materials and Devices (IMD-1), Jülich, Germany

In collaboration with RWTH Aachen University

- **Developed** an electro-mechanical puncture testing methodology for hydrated catalyst coated membranes (CCMs), combining controlled needle indentation with in-situ electrical breakthrough detection to characterize pin-hole initiation under operationally relevant conditions.
- **Quantified** critical loads for delamination and crack initiation across two CCM fabrication routes (doctor-blade-coated, screen-printed) using nanoindentation and microscratch testing.
- **Investigated** mechanical strength of electrode-membrane interfaces under hydrated and non-hydrated conditions, elucidating the role of water uptake in interfacial mechanical response.
- **Implemented** Python-based statistical analysis pipelines (ordinary least squares regression, F-statistics, Pearson correlation, Weibull maximum-likelihood estimation with right-censored run-out data); built automated workflows for cross-experiment comparison and visualization.
- **Performed** post-mortem characterisation of failure sites using scanning electron microscopy and confocal laser scanning microscopy across five voltage conditions, correlating micro-structural features with mechanical and electrical failure behavior.
- **Communicated** findings to international audiences through oral presentation at MSE 2024 (Darmstadt) and through two first-author manuscripts currently in preparation for peer-reviewed journals.
- **Mentored** Master's students in experimental methods, data analysis, and scientific writing.

EDUCATION

PhD, Materials Science

Sep 2022 – 2026 (expected)

Forschungszentrum Jülich GmbH (IMD-1) / RWTH Aachen University, Germany

Dissertation: Multi-method mechanical characterisation of MEAs in PEM fuel cells from indentation to electro-mechanical pin-hole initiation.

Supervisor: Prof. Dr. Ruth Schwaiger.

M.Sc., Applied Physics

2019 – 2021

Amity University, Noida, India • CGPA 9.12 / 10

Thesis: Metal doping in CuS thin films for optical and electronic enhancement.**B.Sc. (Hons.), Physics**

2016 – 2019

Amity University, Noida, India • CGPA 8.01 / 10

CONFERENCE PRESENTATIONS

- **Saxena, A.** (2024). *Nanomechanical failure analysis of thin-film electrodes on polymer membranes for fuel cells*. Oral presentation, MSE Conference 2024, Darmstadt, Germany.
- **Saxena, A.** (2019). *Detecting the topological charge of singular light beams using integral transformation*. Poster presentation, ICRTMD 2019, Noida, India.

TECHNICAL SKILLS

Electrochemical energy systems: PEM fuel cells, PEM electrolyzers, membrane electrode assemblies (MEAs), catalyst coated membranes (CCMs), catalyst coated substrates (CCS), electrode-membrane interfaces.

Mechanical and materials characterisation: Nanoindentation, nanoscratching, thin-film mechanics, interfacial failure analysis, electro-mechanical puncture testing, membrane integrity and pin-hole assessment.

Microscopy and imaging: Scanning electron microscopy (SEM), confocal laser scanning microscopy, optical microscopy, image analysis.

Statistical analysis and programming: Python (NumPy, SciPy, Pandas, Matplotlib), ordinary least squares regression, F-statistics, Pearson correlation, Weibull maximum-likelihood estimation with right-censored data, scientific data visualization, MS Excel.

Professional skills: Scientific writing, oral and poster presentations, experimental design, cross-functional collaboration, project coordination, mentoring.

FELLOWSHIPS & AWARDS

- **HITEC Fellow 2022** – Helmholtz Interdisciplinary Doctoral Training in Energy and Climate Research; competitive fellowship awarded annually to up to eight doctoral researchers conducting climate and energy research at Forschungszentrum Jülich.
- Best in Technical Innovation – M.Sc. Applied Physics, Amity University, India.
- Best in Interpersonal Skills and Team Spirit – M.Sc. Applied Physics, Amity University, India.
- Best in Leadership Qualities – B.Sc. (Hons.) Physics, Amity University, India.

LANGUAGES

English (fluent) • **German** (beginner, A2) • **Hindi** (native)

REFERENCES

Available on request.